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Computer Networks (BCS502) CCA Report

On

“FTP Server Configuration”

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

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Evaluation sheet

Name	USN	Introduction and Setup, Application Selection and Relevance (5)	Packet Capture Process(3)	Packet Analysis and Interpretation(5)	Use of Wireshark Features and Practical Insights and Troubleshooting(5)	Documents and Report	Total
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Introduction

File Transfer Protocol (FTP) is a standard network protocol used to transfer files between a client and a server over a TCP/IP network. In networking education and simulation, Cisco Packet Tracer provides a convenient platform to understand FTP operations without requiring real hardware. This project simulates an FTP server environment to demonstrate file sharing, authentication, and data communication between clients and a server.

Abstract

This report presents the design and simulation of an FTP server setup using Cisco Packet Tracer. The simulation includes the configuration of a server device with FTP services enabled and two PCs accessing the server for uploading and downloading files. The purpose is to understand FTP architecture, commands, and communication mechanisms in a controlled virtual network. The project emphasizes security configurations, testing, and practical applications of FTP in modern networks.

Problem Statement

In most educational and testing environments, setting up an actual FTP server is challenging due to hardware and network constraints. There is a need for a cost-effective, controlled, and safe simulation platform to demonstrate and analyse FTP operations. This project aims to simulate FTP file transfers between a client and a server using Cisco Packet Tracer, ensuring functionality such as login authentication, file accessibility, and secure communication flow.

System Requirements

To create an actual FTP Server configuration in Cisco Packet Tracer, both hardware and software components are required. These components help to build the network, store files, and check the communication between the client and the server.

Hardware Requirements:

- **PC (Client):** Used to access the files from the server.
- **Switch:** Connects all the devices in the network.
- **FTP Server:** Stores and hosts the web page for users to access.
- **Ethernet Cables:** Used to connect the PC, Switch, and Servers together.

Software Requirements:

- **Cisco Packet Tracer:** Software used to design, configure, and simulate the network.
- **Operating System:** Windows, macOS, or Linux for running Cisco Packet Tracer.
- **Text Editor:** To create and open files
- **File Manager:** To manage the files in the PC
- **Command Prompt:** Used to connect to server and execute various ftp commands

Design and Implementation

In this experiment, a small network was designed in **Cisco Packet Tracer** to demonstrate the working of an FTP service.

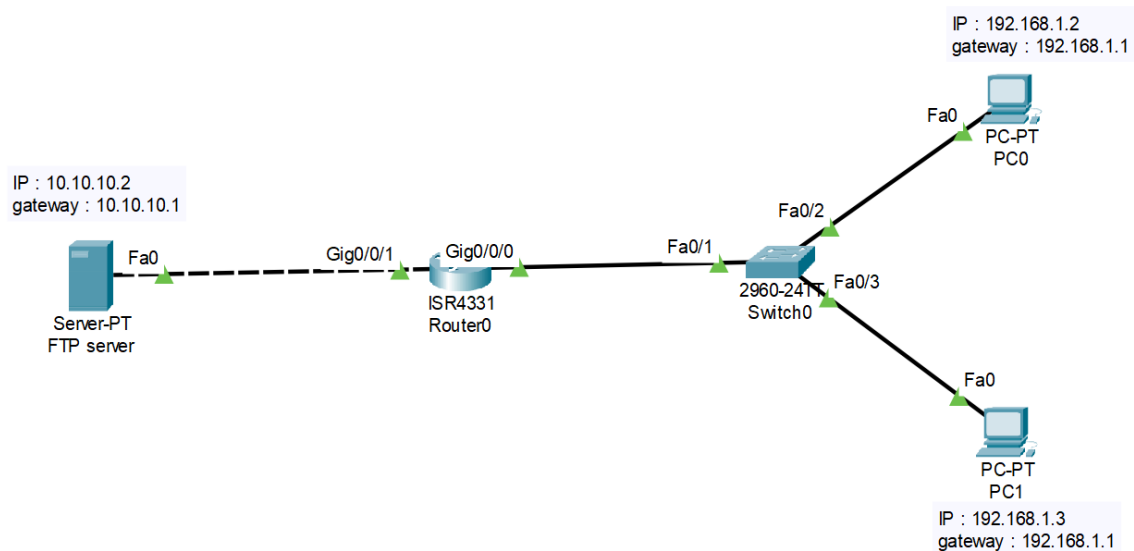
The topology consists of:

- **1 FTP Server** (Server-PT)
- **1 Router** (ISR4331)
- **1 Switch** (2960-24TT)
- **2 Client PCs** (PC0 and PC1)

IP Address Configuration:

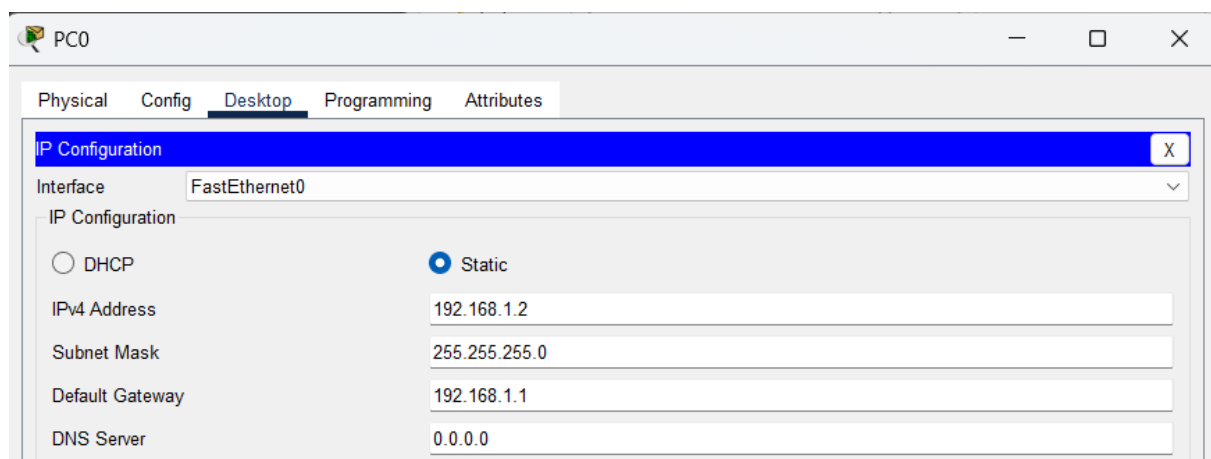
Device	Interface	IP Address	Subnet Mask	Default Gateway
FTP Server	Fa0	10.10.10.2	255.255.255.0	10.10.10.1
Router (Gig0/0/1)	-	10.10.10.1	255.255.255.0	-
Router (Gig0/0/0)	-	192.168.1.1	255.255.255.0	-
PC0	Fa0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	Fa0	192.168.1.3	255.255.255.0	192.168.1.1

They all connected as demonstrated below.

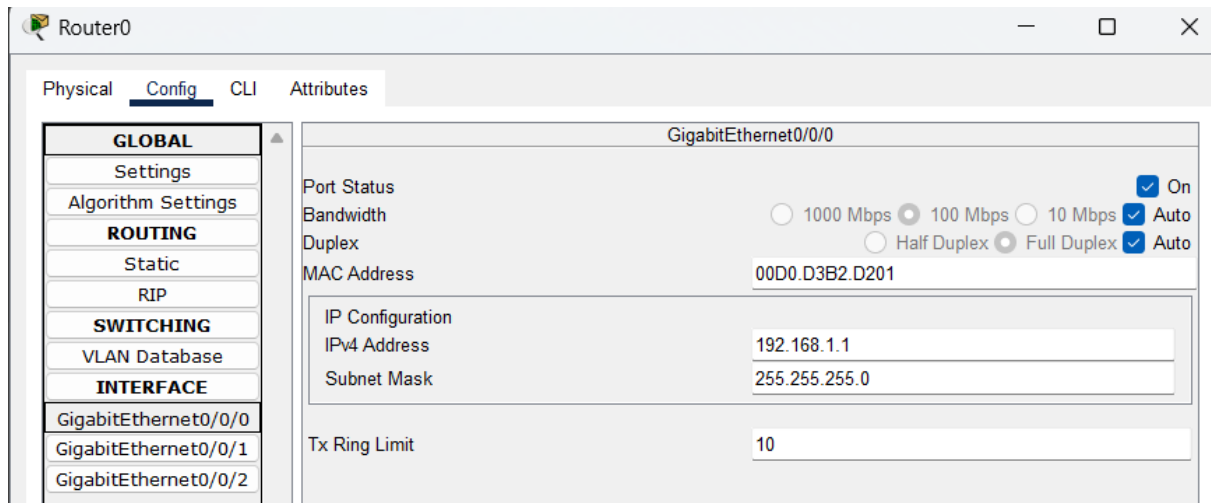


Configuration Steps

1. Set IP Addresses: Assign the listed IPs to each end-device.



2. Router Configuration



The image shows a configuration window for a router named 'Router0'. The 'Config' tab is selected, and the 'GigabitEthernet0/0/0' interface is chosen. The configuration includes:

- Port Status:** On (checked)
- Bandwidth:** 100 Mbps (selected)
- Duplex:** Full Duplex (selected)
- MAC Address:** 00D0.D3B2.D201
- IP Configuration:**
 - IPv4 Address:** 192.168.1.1
 - Subnet Mask:** 255.255.255.0
- Tx Ring Limit:** 10

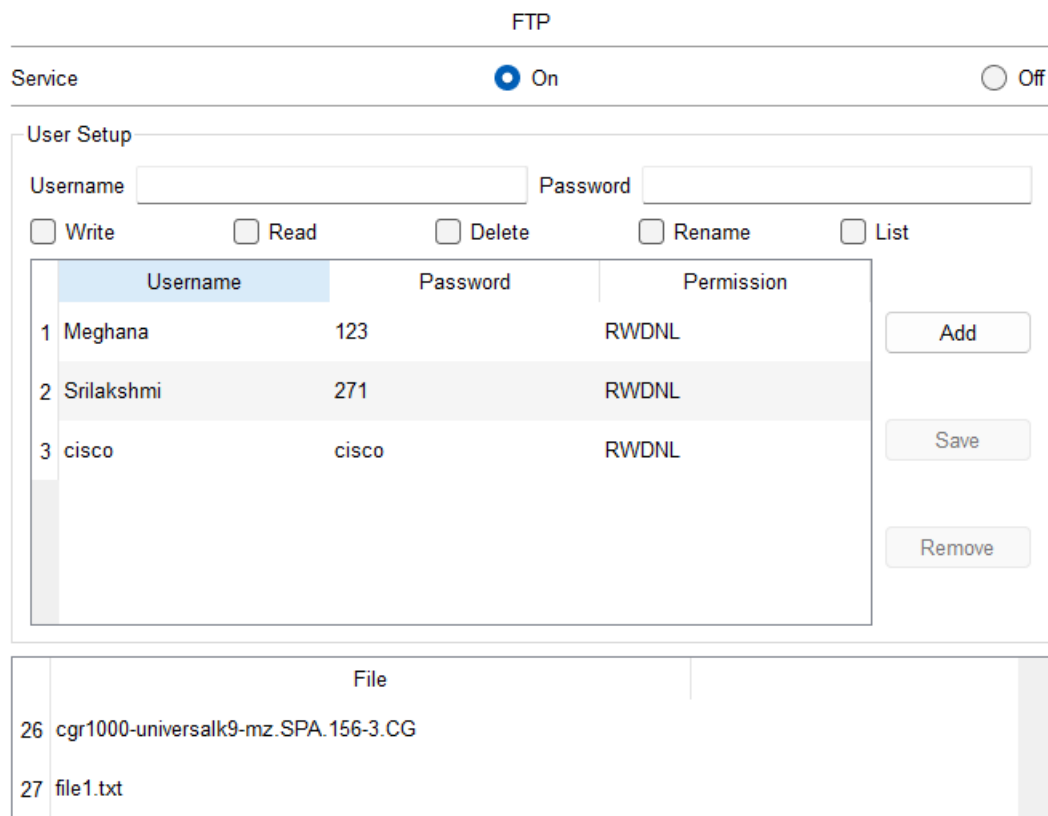
The left sidebar shows a tree view with categories: GLOBAL, ROUTING, SWITCHING, and INTERFACE. Under INTERFACE, the selected interface 'GigabitEthernet0/0/0' is highlighted.

3. FTP Server Configuration

- Go to Services → FTP and enable the FTP service.
- Create two user account, for example:

Username: Meghana

Password: 123



The image shows the 'FTP' configuration window. The 'Service' is set to 'On'. The 'User Setup' section contains a table with user accounts and their permissions.

	Username	Password	Permission
1	Meghana	123	RWDNL
2	Srilakshmi	271	RWDNL
3	cisco	cisco	RWDNL

Buttons: Add, Save, Remove

The bottom section shows a list of files:

	File
26	cgr1000-universalk9-mz.SPA.156-3.CG
27	file1.txt

3. Now lets see what are the files in PC0 and PC1

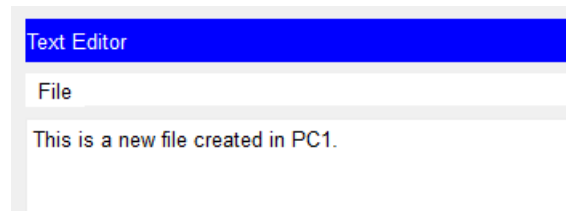
Open command prompt in PC0 and execute

>> **dir** : This shows 2 files in PC1 -file1.txt and sampleFile.txt

Our goal is to transfer this file1.txt from PC1 to PC0 using the FTP server

```
Directory of C:\  
  
1/1/1970    5:30 PM           34      file1.txt  
1/1/1970    5:30 PM           26      sampleFile.txt  
               60 bytes               2 File(s)
```

The content in file1.txt:



Text Editor

File

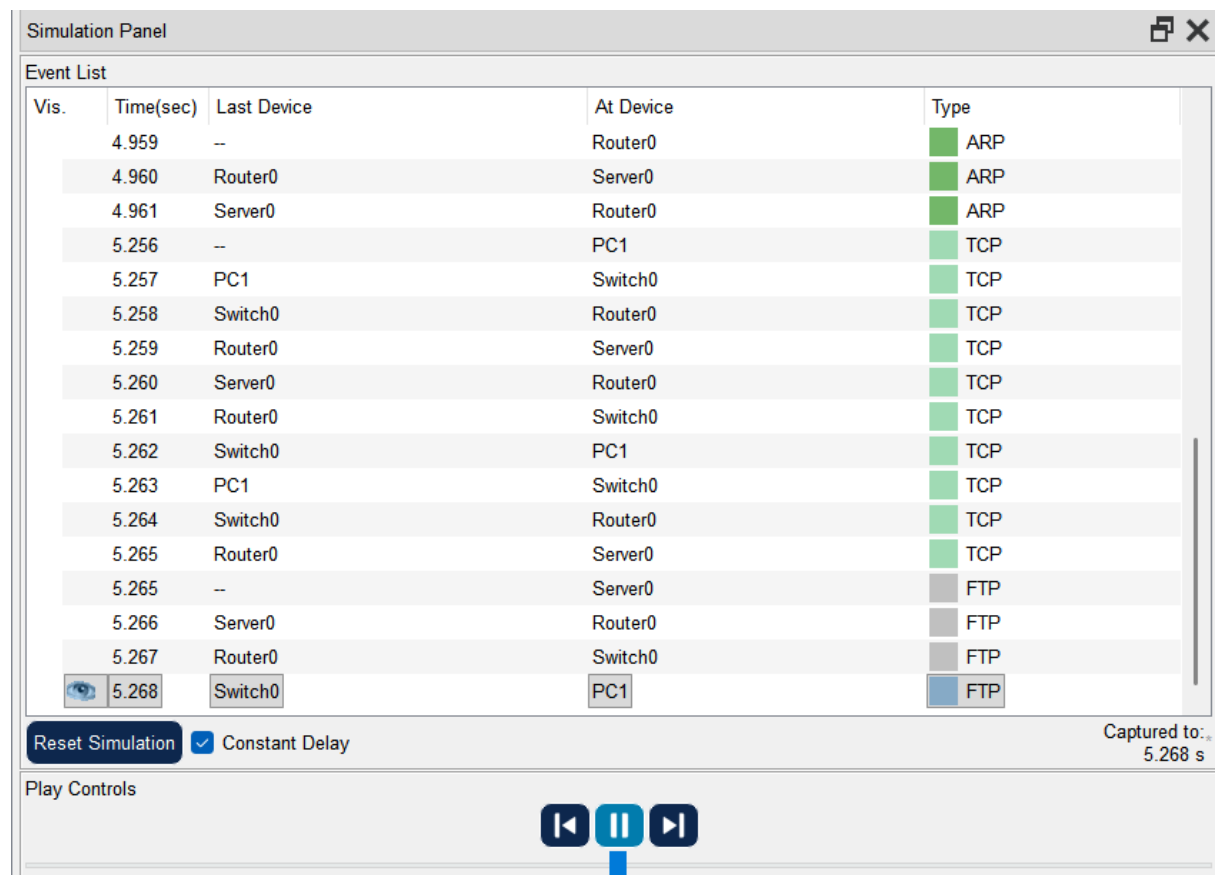
This is a new file created in PC1.

4. Connect to the FTP server using the command

>> **ftp 10.10.10.2**

And login to your user account in PC1

>> **dir** (to see the files in the server)



Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	4.959	--	Router0	ARP
	4.960	Router0	Server0	ARP
	4.961	Server0	Router0	ARP
	5.256	--	PC1	TCP
	5.257	PC1	Switch0	TCP
	5.258	Switch0	Router0	TCP
	5.259	Router0	Server0	TCP
	5.260	Server0	Router0	TCP
	5.261	Router0	Switch0	TCP
	5.262	Switch0	PC1	TCP
	5.263	PC1	Switch0	TCP
	5.264	Switch0	Router0	TCP
	5.265	Router0	Server0	TCP
	5.265	--	Server0	FTP
	5.266	Server0	Router0	FTP
	5.267	Router0	Switch0	FTP
	5.268	Switch0	PC1	FTP

Reset Simulation ☒ Constant Delay

Captured to: 5.268 s

Play Controls


```

C:\>ftp 10.10.10.2
Trying to connect...10.10.10.2
Connected to 10.10.10.2
220- Welcome to PT Ftp server
Username:Meghana
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 10.10.10.2:
0   : asa842-k8.bin                      5571584
1   : asa923-k8.bin                      30468096
2   : cl841-advipservicesk9-mz.124-15.T1.bin 33591768
3   : cl841-ipbase-mz.123-14.T7.bin        13832032
4   : cl841-ipbasek9-mz.124-12.bin        16599160
5   : cl900-universalk9-mz.SPA.155-3.M4a.bin 33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
7   : c2600-i-mz.122-28.bin              5571584
8   : c2600-ipbasek9-mz.124-8.bin         13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin     5571584
12  : c2800nm-ipbasek9-mz.124-8.bin       15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14  : c2950-i6q4l2-mz.121-22.EA4.bin     3058048
15  : c2950-i6q4l2-mz.121-22.EA8.bin     3117390
16  : c2960-lanbase-mz.122-25.FX.bin     4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin    4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin    4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin 33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
26  : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
27  : ir800-universalk9-mz.SPA.155-3.M    61750062
28  : ir800-universalk9-mz.SPA.156-3.M    63753767
29  : ir800_yocto-1.7.2.tar              2877440
30  : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
31  : pt1000-i-mz.122-28.bin              5571584
32  : pt3000-i6q4l2-mz.121-22.EA4.bin     3117390
ftp>|

```

We can not see the file file1.txt in the FTP server

5. To put the file1.txt from the system C drive into the server we use the command

```
>>put file1.txt
```

Now the files in the server are

```

ftp>put file1.txt

Writing file file1.txt to 10.10.10.2:
File transfer in progress...

[Transfer complete - 34 bytes]

34 bytes copied in 0.082 secs (414 bytes/sec)
ftp>dir

Listing /ftp directory from 10.10.10.2:
0   : asa842-k8.bin                      5571584
1   : asa923-k8.bin                      30468096
2   : c184l-advipservicesk9-mz.124-15.T1.bin 33591768
3   : c184l-ipbase-mz.123-14.T7.bin        13832032
4   : c184l-ipbasek9-mz.124-12.bin        16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
7   : c2600-i-mz.122-28.bin              5571584
8   : c2600-ipbasek9-mz.124-8.bin         13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin     5571584
12  : c2800nm-ipbasek9-mz.124-8.bin       15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14  : c2950-i6q412-mz.121-22.EA4.bin     3058048
15  : c2950-i6q412-mz.121-22.EA8.bin     3117390
16  : c2960-lanbase-mz.122-25.FX.bin      4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin    4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin    4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin 33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
26  : file1.txt ← 34
27  : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
28  : ir800-universalk9-mz.SPA.155-3.M    61750062
29  : ir800-universalk9-mz.SPA.156-3.M    63753767
30  : ir800_yocto-1.7.2.tar              2877440
31  : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
32  : pt1000-i-mz.122-28.bin             5571584
33  : pt3000-i6q412-mz.121-22.EA4.bin    3117390
ftp>

```

6. Now we need to transfer this file1.txt to the PC0.

>> **dir** : shows the files in the C drive of PC0 and there is no file1.txt

```

Directory of C:\

1/1/1970    5:30 PM          26      sampleFile.txt
                26 bytes          1 File(s)

```

Connect to the FTP server and login to your account

```

C:\>ftp 10.10.10.2
Trying to connect...10.10.10.2
Connected to 10.10.10.2
220- Welcome to PT Ftp server
Username:Srilakshmi
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 10.10.10.2:
0   : asa842-k8.bin                    5571584
1   : asa923-k8.bin                    30468096
2   : cl841-advipservicesk9-mz.124-15.T1.bin  33591768
3   : cl841-ipbase-mz.123-14.T7.bin    13832032
4   : cl841-ipbasek9-mz.124-12.bin     16599160
5   : cl900-universalk9-mz.SPA.155-3.M4a.bin  33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin  33591768
7   : c2600-i-mz.122-28.bin           5571584
8   : c2600-ipbasek9-mz.124-8.bin      13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin  50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin  33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin    5571584
12  : c2800nm-ipbasek9-mz.124-8.bin     15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin  33591768
14  : c2950-i6q412-mz.121-22.EA4.bin    3058048
15  : c2950-i6q412-mz.121-22.EA8.bin    3117390
16  : c2960-lanbase-mz.122-25.FX.bin     4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin    4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin    4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin  8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin  10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin  33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin  83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin  505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG  159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG  184530138
26  : file1.txt                       34
27  : ir800-universalk9-bundle.SPA.156-3.M.bin  160968869
28  : ir800-universalk9-mz.SPA.155-3.M     61750062
29  : ir800-universalk9-mz.SPA.156-3.M     63753767
30  : ir800_yocto-1.7.2.tar             2877440
31  : ir800_yocto-1.7.2_python-2.7.3.tar  6912000
32  : pt1000-i-mz.122-28.bin            5571584
33  : pt3000-i6q412-mz.121-22.EA4.bin    3117390
ftp>

```

To transfer the file file1.txt from server to PC C drive we use the command

>> **get file1.txt**

And now on C drive of the PC0 we have the file1.txt

```

ftp>get file1.txt

Reading file file1.txt from 10.10.10.2:
File transfer in progress...

[Transfer complete - 34 bytes]

34 bytes copied in 0 secs
ftp>quit

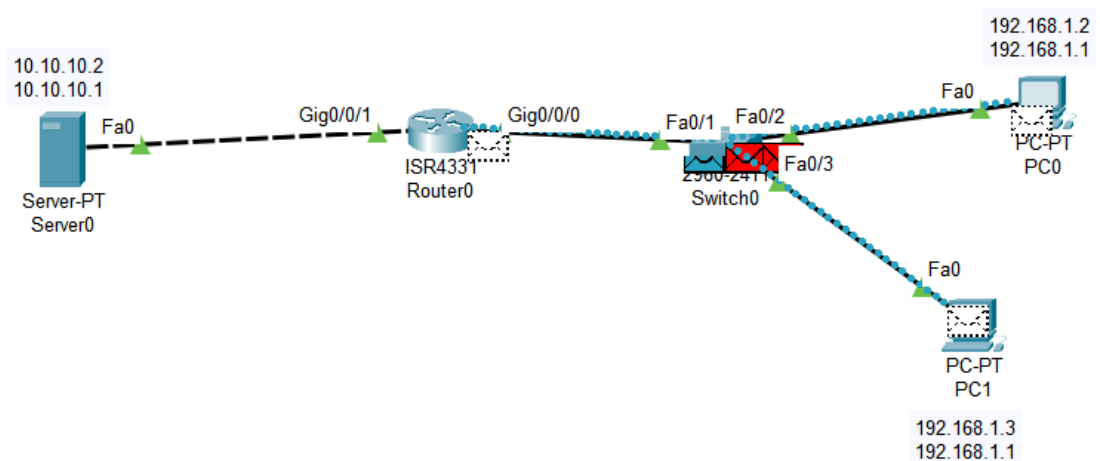
221- Service closing control connection.
C:\>dir

Volume in drive C has no label.
Volume Serial Number is 5E12-4AF3
Directory of C:\

1/1/1970    5:30 PM           34      file1.txt
1/1/1970    5:30 PM           26      sampleFile.txt
              60 bytes           2 File(s)

C:\>

```



Result : Now this file created in PC1 can be accessed from PC0

Security Analysis

The FTP protocol transmits data in plaintext, making it insecure for sensitive transfers. To enhance security:

- Replace FTP with **SFTP** or **FTPS** (encrypted versions).
- Use **strong authentication credentials**.
- Restrict FTP access through **Access Control Lists (ACLs)** on the router.
- Monitor network traffic using **Packet Tracer's simulation mode** or real-world tools like Wireshark.

System Testing

The following tests were performed after setting up the network

- Tests were performed to confirm:
- **Connectivity** between all devices using ping commands.
- **FTP accessibility** by connecting to the server using both PC0 and PC1.
- **File operations** such as upload and download were successful.
- **Routing functionality** allowed devices on different subnets to communicate.

All tests were successful, confirming that the FTP setup worked as intended.

Innovation and Application relevance

The FTP server simulation can be expanded with innovative extensions and real-world applications beyond basic file transfer operations. The following ideas demonstrate how this setup can be developed further:

1. Implement **Access Control Lists (ACLs)** on routers to allow FTP access only from specific clients or subnets.

2. Use **Python scripts** or **Cisco Embedded Event Manager (EEM)** to automate FTP file uploads or backups.
3. Compare FTP with **TFTP, SFTP, and SCP** in terms of performance, speed, and security.

Conclusion

The FTP Server Simulation using Cisco Packet Tracer was successfully designed and implemented. Communication between clients and the FTP server across different networks was established through router configuration. This project helped in understanding key networking concepts such as IP addressing, routing, and service configuration. It also highlighted the importance of securing file transfer services in real-world networks.

Reference

- Cisco Networking Academy, “Introduction to Packet Tracer”, Cisco Systems.
- Cisco Packet Tracer Official Documentation – <https://www.netacad.com/>
- TutorialsPoint, “HTTP Protocol and Web Server Basics” – <https://www.tutorialspoint.com/>
- GeeksforGeeks, “DNS Server and Its Working” – <https://www.geeksforgeeks.org/>
- StudyTonight, “Client-Server Architecture and Web Communication” – <https://www.studytonight.com/>